

AI2200: CONCENTRATION INEQUALITIES

PRACTICE PROBLEMS



TOPICS: Sub-Gaussian Inequalities, Sub-Exponential Inequalities

Recall that a random variable X is said to be σ -sub-Gaussian if

$$\mathbb{E} \left[\exp(t(X - \mathbb{E}[X])) \right] \leq \exp\left(\frac{t^2 \sigma^2}{2}\right) \quad \forall t \in \mathbb{R}.$$

1. Fix $a, b \in \mathbb{R}$ with $a < b$.

(a) Suppose that X is a random variable taking values in the bounded interval $[a, b]$. Show that

$$\text{Var}(X) \leq \frac{(b-a)^2}{4}.$$

For what distribution of X is equality achieved in the above inequality?

(b) Suppose that X is a continuous random variable with zero mean, taking values in the bounded interval $[a, b]$. Let the PDF of X be denoted by f . For a fixed $\lambda \in \mathbb{R}$, let Z_λ be a random variable whose PDF g_λ is defined as

$$g_\lambda(x) = \frac{1}{M_X(\lambda)} \exp(\lambda x) f(x), \quad x \in [a, b],$$

where M_X denotes the moment-generating function of X . Show that

$$\frac{d^2}{d\lambda^2} M_X(\lambda) = \text{Var}(Z_\lambda).$$

(c) If X is a random variable with $\mathbb{E}[X] = 0$ and taking values in the interval $[a, b]$, show that X is σ -sub-Gaussian with $\sigma = \frac{b-a}{2}$.

Hint: Define the log-moment generating function of X , ψ_X as

$$\psi_X(\lambda) = \log M_X(\lambda), \quad \lambda \in \mathbb{R}.$$

Using the Taylor expansion for ψ_X around $\lambda = 0$, we get

$$\psi_X(\lambda) \leq \psi_X(0) + \psi'_X(0)\lambda + \psi''_X(\lambda') \frac{\lambda^2}{2} \quad \text{for some } \lambda' \in [0, \lambda].$$

Evaluate $\psi_X(0)$, $\psi'_X(0)$ (by freely passing derivatives inside expectations if needed), and use the fact that

$$\psi''_X(\lambda') = \text{Var}(Z_{\lambda'}) \stackrel{\text{part (a)}}{\leq} \frac{(b-a)^2}{4}.$$

(d) Suppose that $X_1, \dots, X_n$ are independent (not necessarily identically distributed), with X_k taking values in the bounded interval $[a_k, b_k]$ with $a_k < b_k$ for each k . Prove that

$$\mathbb{P} \left(\left\{ \frac{1}{n} \sum_{k=1}^n (X_k - \mathbb{E}[X_k]) \geq \varepsilon \right\} \right) \leq \exp \left(\frac{-2n^2 \varepsilon^2}{\sum_{k=1}^n (b_k - a_k)^2} \right) \quad \forall \varepsilon > 0.$$

2. Let $X_1, \dots, X_n$ be independent (not necessarily identically distributed) random variables with

$$X_k - \mathbb{E}[X_k] \leq b \quad \text{with probability 1} \quad \forall k \in \{1, \dots, n\},$$

for some fixed $b \in \mathbb{R}$. Let S and V be defined as

$$S := \sum_{k=1}^n (X_k - \mathbb{E}[X_k]), \quad V := \sum_{k=1}^n \text{Var}(X_k).$$

(a) Show that the function $g : \mathbb{R} \rightarrow \mathbb{R}$ defined as

$$g(x) = \frac{\exp(x) - x - 1}{x^2}, \quad x \in \mathbb{R},$$

is increasing.

(b) If X is a random variable with $\mathbb{E}[X] = 0$ and $X \leq b$ with probability 1, show that

$$\mathbb{E}[\exp(X)] \leq 1 + g(b) \text{Var}(X).$$

(c) Prove that

$$(1 + \alpha) \log(1 + \alpha) - \alpha \geq \frac{3\alpha^2}{6 + 2\alpha} \quad \forall \alpha \geq 0.$$

(d) For any $\varepsilon > 0$ and $\alpha = \frac{b\varepsilon}{V}$, show that

$$\begin{aligned} \mathbb{P}(\{S \geq \varepsilon\}) &\leq \exp\left(-\frac{V}{b^2} ((1 + \alpha) \log(1 + \alpha) - \alpha)\right) \\ &\leq \exp\left(\frac{-\varepsilon^2}{2V \left(1 + \frac{b\varepsilon}{3V}\right)}\right). \end{aligned}$$

(e) Using the result of part (d), show that for any $\delta > 0$,

$$\mathbb{P}\left(\left\{S \geq \sqrt{2V \log\left(\frac{1}{\delta}\right)} + \frac{2B}{3} \log\left(\frac{1}{\delta}\right)\right\}\right) \leq \delta.$$

3. Let $X \sim \text{Ber}(p)$.

(a) Show that X is $(1/2)$ -sub-Gaussian for any $p \in [0, 1]$.

(b) The **sub-Gaussianity constant** of any sub-Gaussian random variable Y is defined as

$$\text{SG}(Y) := \inf \left\{ K > 0 : \mathbb{E}[\exp(t(X - \mathbb{E}[X]))] \leq \exp\left(\frac{t^2 K^2}{2}\right) \quad \forall t \in \mathbb{R} \right\}$$

Show that for any $p \in (0, 1)$, the sub-Gaussianity constant of $X \sim \text{Ber}(p)$ is given by

$$Q(p) = \sqrt{\frac{1 - 2p}{2 \log\left(\frac{1-p}{p}\right)}}.$$

(c) Compute the value of $Q(p)$ for $p = \frac{1}{2}$.

4. Let X be a discrete random variable with the PMF

$$p_X(k) = \frac{6}{\pi^2 k^2}, \quad k \in \mathbb{N}.$$

- (a) What is $\mathbb{E}[X]$?
 (b) Is X sub-Gaussian or sub-Exponential? Justify.

Hint: Use the fact that

$$\sum_{k=n}^{\infty} \frac{1}{k^2} \geq \int_n^{\infty} \frac{1}{x^2} dx \quad \forall n \in \mathbb{N}.$$

5. Let $X_1, \dots, X_n$ be σ -sub-Gaussian random variables (not necessarily independent), and let $Z = \max\{X_1, \dots, X_n\}$.

- (a) Show that

$$\exp(\lambda \mathbb{E}[Z]) \leq n \exp\left(\frac{\lambda^2 \sigma^2}{2}\right) \quad \forall \lambda \in \mathbb{R}.$$

Hint: $\exp(\lambda \mathbb{E}[Z]) \leq \mathbb{E}[\exp(\lambda Z)]$, courtesy of Jensen's inequality.

- (b) Optimise the result in part (a) over λ to show that

$$\mathbb{E}[Z] \leq \sqrt{2\sigma^2 \log n}.$$

- (c) Show that

$$\mathbb{P}\left(\left\{Z \geq \sqrt{2\sigma^2 \log \frac{n}{\delta}}\right\}\right) \leq \delta \quad \forall \delta \in (0, 1).$$